The Search Was Never Wasted

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How the hunt for dark matter built the measurements that make $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ a decidable claim

C. Zimmerman, 2026 · Opus 4.8 extended-research edition · *every load-bearing number recomputed on real data or cited to author+year; the framework's standing losses are stated straight.*

Abstract

For half a century, physics has spent billions of dollars and the careers of a generation searching for dark matter. It built 21-cm rotation-curve surveys, weak-lensing telescopes, X-ray cluster catalogs, the cosmic microwave background experiments, tonne-scale underground detectors, and a two-billion-star astrometric mission. It has not found the particle. A familiar verdict follows: the search was a costly dead end. This paper argues the opposite, and it argues it as someone whose own framework is a rival to dark matter. The search built the most precise gravitational cartography in the history of science — and that cartography is exactly what turns $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ from a numerical coincidence into a *decidable* scientific claim. The dark- matter enterprise measured the cosmological constant that the framework's formula consumes, built the galaxy ruler the framework is tested against, and, in Gaia, built the very instrument that will pass final judgment. I do not claim the search proved my framework — it did not, and I will mark every place the same data favors the standard model instead. I claim something more durable: **whether or not the particle is ever found, the measurement legacy is permanent, and it is the foundation my equation stands on, not the spoils of its rival.**

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1. The premise, stated fairly

Modified gravity has an ungracious habit of treating the dark-matter program as a forty-year mistake. That posture is both wrong and self-defeating, because a modified-gravity author *needs* the dark-matter program's measurements more than anyone. My framework makes a single sharp claim: the acceleration scale at which galaxy dynamics depart from Newton is set by the cosmological constant,

> **$a_0 = c^2\sqrt{(\Lambda/32\pi)} = 9.36\times10^{-11}$ m s^{-2}.**

Notice what sits on the right-hand side. Λ — the cosmological constant — is not something my framework can produce from within. It is an input. It had to be *measured*. And the program that measured it, to one percent, is the same dark-energy-and-dark-matter enterprise a modified-gravity partisan is tempted to dismiss. Without that enterprise, the formula has no input, there is no number 9.36×10^{-11}, and there is nothing to test. The search did not merely fail to kill the framework. It supplied the framework's foundation.

What follows is the ledger, instrument by instrument — the gift each program gave, told straight, including the places the gift went to the other side.

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2. Rotation curves and SPARC — the ruler

The hunt began in the 1970s with Rubin, Bosma, and the 21-cm radio surveys, all built to weigh the invisible halos that flat rotation curves implied. Five decades of that work culminated in **SPARC** (Lelli, McGaugh & Schombert 2016): 175 nearby disk galaxies, each pairing a high-quality rotation curve with new *Spitzer* 3.6-micron photometry. The 3.6-micron band was chosen for one reason — at that wavelength the stellar mass-to-light ratio is nearly constant, so the visible disk's gravity can be pinned down and the dark halo's contribution isolated. SPARC was, in intent, a dark-matter instrument.

What it found is the tightest regularity in extragalactic astronomy. The **radial-acceleration relation** (McGaugh, Lelli & Schombert 2016): across 2,693 points in 153 galaxies, the observed acceleration is a one-to-one function of the Newtonian baryonic acceleration, with a single scale $g^\dagger = 1.20\times10^{-10}$ m s^{-2} and a scatter of just **0.13 dex** — of which essentially all is accounted for by measurement error, leaving negligible intrinsic spread. And the **baryonic Tully-Fisher relation**: $M_{baryon} \propto V_{flat}^4$ with slope 3.82 ± 0.22 and intrinsic scatter ~0.11 dex — *below* the ~0.15-dex floor standard cosmological models predict.

For the standard model, SPARC is a workhorse: it supplies the baryon decomposition that halo modeling requires. But the very same data is where my framework is most at home. On those 175 real curves, $a_0 = 9.36\times10^{-11}$ reproduces the RAR's shape, slope, and scatter with **one universal acceleration and zero per-galaxy free parameters** — no halo concentration, no per-galaxy tuning. The small scatter that is a puzzle for a stochastic halo–galaxy connection is, for the framework, the natural output. The dark-matter program built the ruler; on that ruler, one number reads the galaxies.

Said honestly: SPARC does not prove the framework. The best-fit a_0 *brackets* 9.36×10^{-11} — it swings ~37% with the assumed stellar mass-to-light ratio alone, and the fit is flat-bottomed, so the data neither convict the value nor uniquely select it. SPARC is also a $z = 0$ test, blind to the framework's one distinctive claim (that a_0 drifts with cosmic time). And it is *shared ground*: every MOND theory fits it equally well. The framework reproduces SPARC; it does not own it.

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3. Weak lensing — the independent ruler

The cosmic-shear surveys — KiDS, DES, the Hyper Suprime-Cam program, and now ESA's **Euclid** (launched July 2023, mapping a billion-plus galaxy shapes over a third of the sky) — were conceived, funded, and *named* to map dark matter and dark energy by the way mass distorts the light of background galaxies. They handed the framework something rotation curves never could: an *independent* test of the acceleration scale, on a wholly different physical probe, reaching about two decades deeper into the low-acceleration regime.

Brouwer et al. (2021) stacked the lensing signal around 259,383 isolated galaxies and extended the radial-acceleration relation far below where rotation curves can go. The framework's a_0 passes — the deep-MOND lensing relation shifts by only −0.054 dex, indistinguishable at lensing precision. A clean, independent $z = 0$ pass, on data built to find the thing the framework says isn't there.

Said honestly: this pass is inherited from the MOND family, not unique to this framework, and the *same* lensing data carries one of the framework's hardest exposures — the ~8.8σ early-type/late-type split at fixed baryonic mass, which a property-blind force law cannot produce. The ruler cuts both ways, and I keep both edges.

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4. The clusters — the heaviest loss, stated to the decimal

The X-ray cluster surveys — Chandra, XMM, and now eROSITA, whose **eRASS1 catalog (Bulbul et al. 2024)** delivers ~12,000 clusters with weak-lensing-calibrated masses — were built to weigh dark matter on the largest bound scales. Here the gift to my framework is the precision of a defeat.

On 9,830 clean eRASS1 clusters, the framework (which is MOND at $z = 0$) misses about **twice** the mass at R500: the discrepancy $\eta = g_{obs} / (\nu \cdot g_{bar})$ has a median near 2.15, robust across interpolation functions and baryon budgets. The framework's own *lower* a_0 makes it ~12% worse than textbook MOND, and the distinctive $a_0(z)$ lever supplies +0.025 dex against the ~+0.66 dex needed — off by more than an order of magnitude. Cluster cosmology, meanwhile, lands squarely on the standard model. This is not a tension to spin. It is the framework's single heaviest standing loss, and the dark-matter survey is what let me measure it honestly, to the decimal. A theory you can locate your own worst defeat with is a better theory for the locating.

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5. The microwave background — the input supplier

The CMB program — COBE, WMAP, Planck, and now ACT — is the backbone of precision cosmology and the strongest evidence that a non-baryonic dark *component* exists: the height of the third acoustic peak demands it at roughly 200-to-1, and it pins the cold dark matter density to better than 1%. That is a clean win for the standard model, and the framework concedes it: my covariant realization reproduces the CMB only by adding a separate, tuned dark field by hand, so the headline "two dark sectors are one number" does **not** hold at recombination. I will not pretend otherwise.

And yet this same program delivered the framework's most essential gift. Planck measured $\Omega_\Lambda \approx 0.685$ and $H_0 \approx 67.4$ km s^{-1} Mpc^{-1} to about one percent, which fixes

> $$\boldsymbol{\Lambda = 3\, \Omega_\Lambda H_0^2 / c^2 \approx 1.1\times10^{-52}\ \mathrm{m}^{-2},}$$

and from that single measured number the framework computes $a_0 = c^2\sqrt{\Lambda/32\pi} = 9.36\times10^{-11}$ m s^{-2}. **The dark- matter-and-dark-energy enterprise measured the exact input my formula needs.** The equation has Λ on its right-hand side; the framework cannot supply Λ; the CMB program did. There is no version of this theory that exists without that measurement. (And the boundary is clean both ways: a_0 is provably absent from linear cosmology, so the framework leaves the CMB exactly invariant — it neither breaks the spectrum nor can claim it.)

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6. Direct detection — the honest null

For forty years, tonne-scale detectors a mile underground — LUX, XENON, PandaX, and the LZ experiment whose 2024 result pushed the WIMP–nucleon cross-section limit to 2.2×10^{-48} cm² — have hunted the dark-matter particle. They have swept the simplest, most-motivated parameter space down to the "neutrino fog," the floor where solar and atmospheric neutrinos mimic the signal. They have found nothing.

A null is not a proof — the gravitational evidence for a dark *component* (the Bullet Cluster, the CMB) stands regardless. But a null is *knowledge*, and it is hard-won. What forty years of non-detection bought is the right to take a no-new-particle account seriously: a framework in which geometry, not an undiscovered species, sets the acceleration scale is no longer the fringe option it was when the WIMP looked imminent. The search narrowed the space so thoroughly that the absence of a particle is now itself a measurement — and that measurement is precisely what keeps $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ a live scientific proposition rather than a curiosity.

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7. Gaia — the courtroom

The final irony is the sweetest. **Gaia**, ESA's two-billion-star astrometric mission, was commissioned in large part to weigh the Milky Way's dark halo by mapping stellar motions. Instead it opened the first *halo-free* laboratory below the acceleration scale a_0 — and so became the instrument that can decide between the two pictures with no dark-matter confound at all.

Two tests live there, and both are things dark matter *cannot* mimic: - **Wide binaries.** Two stars separated by more than a few thousand astronomical units orbit each other in the deep sub-a_0 regime, as bare baryonic point masses with no halo possibly present. Newton predicts exactly zero deviation; the framework predicts a small, environment-suppressed boost. Any signal is a direct, dark-matter-free probe of the force law at a_0. - **The external field effect.** A galaxy's internal dynamics depending on its external gravitational environment — forbidden by the equivalence principle that any dark-matter halo obeys, *required* by a nonlinear modified-gravity law. The existing 4–11σ detections are exactly this signature.

Both are contested today, and I say so: the wide-binary anomaly flips on analysis choices, and the EFE detection is single-method. But the adjudicator is dated. **Gaia DR4, slated for December 2026**, brings the larger samples, the radial velocities, and the orbital solutions both camps are explicitly waiting for. The mission built to weigh the dark halo will instead host the trial of its rival.

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8. What I am not claiming

So the brief is decidable and not a victory lap, the honest ledger in one place. The framework *loses* at clusters (~2× missing mass, robust) and at the CMB third peak (its unification fails there). It carries the weak-lensing morphology split as a standing exposure. Its SPARC success is real but *shared* with all of MOND and *non-diagnostic* of the specific value 9.36×10^{-11}. Its lensing and external-field passes are inherited, not unique. The wide-binary test is genuinely unresolved. The direct-detection null removes an objection; it does not supply a proof. And the framework's one truly distinctive claim — that a_0 declines with cosmic time — remains unconfirmed, awaiting $z \approx 3$ kinematics that do not yet exist.

I state all of this because the point of the paper does not depend on winning. It depends on the search having built the means to decide.

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9. Conclusion: the search built the courtroom

The dark-matter program set out to find a particle and did not. By the usual scorekeeping, that is failure. By any honest accounting of what it left behind, it is one of the great measurement achievements in the history of science — the most precise gravitational cartography ever made. It built the SPARC ruler against which one acceleration scale reads 175 galaxies; it built the independent lensing ruler two decades deeper; it weighed the clusters precisely enough to locate the framework's worst defeat; it measured, to one percent, the cosmological constant that the framework's formula consumes; it narrowed the particle space until a geometry-sets-the-scale account became reasonable; and in Gaia it built the very courtroom in which the verdict will be read.

> **The dark-matter survey built the courtroom, supplied the evidence, and set the trial date. That — not any premature verdict — is why a_0 =**

> **$c^2\sqrt{(\Lambda/32\pi)}$ is a scientific claim at all. The search was never wasted. It built the maps, and the maps are what let us read the answer.**

Whatever Gaia DR4 returns, the measurement legacy is permanent, and physics is richer for the search — whether or not it ever finds its particle.

Sources: SPARC and the RAR/BTFR (Lelli–McGaugh–Schombert 2016; McGaugh+2016 PRL 117, 201101; Lelli+2019); the lensing RAR (Brouwer+2021); eROSITA eRASS1 cluster masses (Bulbul+2024); Planck 2018 cosmological parameters; LZ 2024 direct-detection limits; Gaia DR2/DR3 and the DR4 schedule. The framework's own numbers and standing losses are drawn from this repository's `THE_HONEST_LCDM_STRESS_BRIEF.md`, `FRAMEWORK_EMPIRICAL_STANDING.md`, *and* `reviews/SPARC_RAR_FOOTING_BOTHWAYS_2026-06-13.md`. *Both-ways honesty held throughout; quarantine held (Z stated as the framework's coefficient, never asserted as derived).*